RAG Assignment: Baseline vs Cross-encoder Re-ranking

I Puneeth Abhishek | 2026701051

GitHub link : https://github.com/Puneeth-Abhishek-6622/rag-baseline-reranking.git

Abstract

This experiment compares dense bi-encoder retrieval with a retrieve-then-rerank pipeline on the D&D Basic Rules PDF. The 180-page document was split into 1,008 page-bounded chunks. Chunks were embedded using sentence-transformers/all-MiniLM-L6-v2 and indexed with FAISS. For each query, dense retrieval returned ten candidates. The baseline selected the top three directly; the advanced pipeline re-scored the same ten query–passage pairs using cross-encoder/ms-marco-MiniLM-L-6-v2 and selected the top three re-ranked chunks. Across six answerable evaluation questions, mean Recall@3 increased from 0.7222 to 0.9444 and mean MRR@10 increased from 0.6667 to 0.9167. Re-ranking improved the ordering of relevant evidence but added latency.

1. Objective and setup

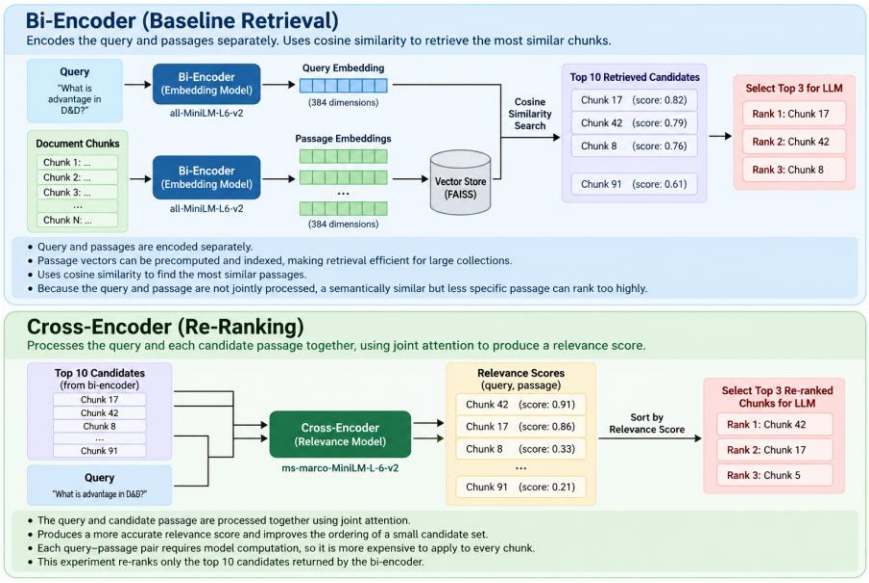
The objective was to compare two retrieval workflows over the same chunked corpus and use the selected passages to generate answers with the same prompt structure. The source was D&D Basic Rules, Version 1.0, released November 2018. Text was extracted from 180 pages and divided into 1,008 chunks while retaining page metadata; chunks were kept within page boundaries so retrieved evidence could be traced to PDF pages.

Each chunk was encoded independently with sentence-transformers/all-MiniLM-L6-v2 (384-dimensional embeddings). The vectors were L2-normalized, so inner-product search is equivalent to cosine similarity. FAISS retrieved the top ten candidates. The cross-encoder then scored each query–candidate pair jointly and sorted out those ten candidates. Both workflows supplied three chunks to the answer-generation function.

The evaluation used six answerable questions, Q1–Q6, with manually annotated supporting pages. An additional query about a weapon property from the 2024 Player's Handbook tested out-of-document answerability and was excluded from metric averages.

2. Architecture comparison

The system uses a two-stage retrieval pipeline. A bi-encoder (all-MiniLM-L6-v2) encodes queries and document chunks separately, retrieving the top 10 candidates using FAISS and cosine similarity. A cross-encoder (ms-marco-MiniLM-L-6-v2) then scores each candidate jointly with the query and re-ranks them, selecting the top three chunks for answer generation. This combines efficient retrieval with improved relevance ranking at a manageable computational cost.



3. Quantitative results

Metrics are averaged over six answerable queries. Recall@k is page-level: a supporting page is counted when at least one chunk from that page appears in the top k. MRR@10 is the mean reciprocal rank of the first relevant result. Hit@3 indicates whether at least one supporting page appears in the final top three.

Metric	Dense baseline	Dense + cross encoder
Mean Recall@1	0.3889	0.7222
Mean Recall@3	0.7222	0.9444
Mean Recall@5	0.7222	1.0000
Mean Recall@10	1.0000	1.0000
Mean MRR@10	0.6667	0.9167
Mean Hit@3	0.8333	1.0000

4. Failure-mode and rank-shift analysis

Q5 — comparison/distractor case: chunk_00065 on PDF page 11, containing the melee/ranged attack ability-modifier rule and finesse exception, moved from dense rank 6 to re-ranked position 2. The baseline top-three pages were 76, 48, and 49, giving Recall@3 = 0.0000. The re-ranked top-three pages were 76, 11, and 49, giving Recall@3 = 1.0000. The relevant passage was in the initial top ten but outside the baseline answer context; re-ranking promoted it into the final context.

Q6 — cross-section/procedural detail case: chunk_00054 on PDF page 9 moved from dense rank 8 to re-ranked position 2. The baseline top-three pages were 70, 70, and 27; the re-ranked top-three pages were 70, 9, and 70. Recall@3 increased from 0.3333 to 0.6667. This is partial evidence recovery, not a complete baseline failure, because the baseline already retrieved relevant evidence from page 70.

These cases illustrate that re-ranking can promote relevant evidence that the first-stage retriever found but ranked too low. It cannot recover a relevant passage that is absent from the initial top ten.

Query	Chunk	Page	Dense rank	Re-ranked	Recall@3 baseline → reranked
Q5	chunk_00065	11	6	2	0.0000 → 1.0000
Q6	chunk_00054	9	8	2	0.3333 → 0.6667

5. LLM answer comparison

The notebook generated Answer 1 and Answer 2 for Q5 and Q6 using the same generation function but different top-three contexts. For Q5, the baseline context did not include page 11, while the re-ranked context did. The re-ranked answer explicitly describes Strength for melee weapon attacks, Dexterity for ranged weapon attacks, and the finesse exception, matching the newly included passage.

For Q6, both contexts include page-70 material, while the re-ranked context also includes page 9. This gives the answer generator additional relevant context, but retrieval metrics alone do not prove every generated statement is more accurate. The generated text and citations must be checked against the source PDF. The notebook's answer_review_sheet.csv is still a human-review worksheet; it should be completed after checking support, completeness, and unsupported claims. This report therefore does not claim a verified reduction in hallucinations.

6. Latency profiling

Latency was measured separately for dense query retrieval and cross-encoder re-ranking. The notebook ran on CPU. These figures are specific to this run and are not universal performance guarantees.

Operation	Mean (s)	Median (s)	Std. dev. (s)
Dense query + FAISS retrieval	0.0088	0.0068	0.0062
Cross-encoder re-ranking (10)	0.1521	0.1348	0.0324
Combined retrieval + re-ranking	0.1610	0.1479	0.0319

7. Scaling question

A cross-encoder processes each query together with each candidate passage. Scoring all 1,008 chunks would require 1,008 pairwise scoring operations per query, compared with ten in the re-ranking stage. Dense retrieval first narrows the search space; the cross-encoder then spends additional computation on a small candidate set. This retrieve-then-rerank architecture balances search efficiency and relevance, but its quality depends on relevant passages entering the initial candidate pool.

8. Limitations

The evaluation contains only six answerable questions, so the results are a small case study rather than a general guarantee. Supporting pages were manually annotated, and page-level recall does not establish that a passage contains every detail needed for a complete answer. Latency depends on hardware, model state, software versions, and input length. Cross-encoder scores are ranking signals, not probabilities of correctness. The generated answers need human review before drawing strong conclusions about factuality.

9. Conclusion

In this experiment, re-ranking improved the average ordering of relevant evidence. Mean Recall@3 increased from 0.7222 to 0.9444, and mean MRR@10 increased from 0.6667 to 0.9167. In Q5, the supporting passage moved from rank 6 to rank 2 and Recall@3 increased from 0 to 1. In Q6, a supporting passage moved from rank 8 to rank 2 and Recall@3 increased from 0.3333 to 0.6667. The gain came with additional mean retrieval latency: about 8.8 ms for dense retrieval versus 161.0 ms for the combined pipeline. The results support using a cross-encoder as a second-stage ranker when relevant passages are retrieved but not ranked highly enough for the final context, subject to the small evaluation size and the need for answer-level review.